Homework 03

Consider the problem of monthly production planning under demand uncertainty for a manufacturer that produces three products: *Product-a*, *Product-b*, and *Product-c*. The manufacture of each of these products requires raw material, *Material-abc*, and three manufacturing processes: *Process-1*, *Process-2*, and *Process-3*. The production process uses machines for each process and thus takes time and incurs costs. In this setting involving demand uncertainty, the materials and/or products and the processing time have to be acquired/manufactured *before* product demand becomes known. In other words, the resource acquisition decisions have to be made here-and-now before demand is realized. The amounts of each resource needed, cost, and maximum amount that can be acquired to make each of the products are given in Table 1. The table also shows the *testing and inspection cost* for each product after it is made and the planned demand amounts that are used in a deterministic setting.

First-stage Decision Variables:

- x_1: Number of units of material acquired.
- x_2: Number of Process-1 hours acquired.
- x_3: Number of Process-2 hours acquired.
- x_4: Number of Process-3 hours acquired.

Second-stage Decision Variables:

- y_1: Number of Product-a produced.
- y_2: Number of Product-b produced.
- y_3: Number of Product-c produced.

Deterministic Parameters:

- m_a: Units of Material-abc needed to produce a unit of Product-a.
- m_b: Units of Material-abc needed to produce a unit of Product-b.
- m_c: Units of Material-abc needed to produce a unit of Product-c.
- m: Units of Material-abc available.
- c_1: Cost of Material-abc (\$/unit).
- h_{1a}: Hours of Process-1 needed to produce a unit of Product-a.
- h_{1b}: Hours of Process-1 needed to produce a unit of Product-b.
- h_{1c}: Hours of Process-1 needed to produce a unit of Product-c.
- h_{2a}: Hours of Process-2 needed to produce a unit of Product-a.
- h_{2b}: Hours of Process-2 needed to produce a unit of Product-b.
- h_{2c}: Hours of Process-2 needed to produce a unit of Product-c.
- h_{3a}: Hours of Process-3 needed to produce a unit of Product-a.
- h_{3b}: Hours of Process-3 needed to produce a unit of Product-b.
- h_{3c}: Hours of Process-3 needed to produce a unit of Product-c.
- h_1: Hours of Process-1 available.
- h_2: Hours of Process-2 available.
- h_3: Hours of Process-3 available.
- c_2: Unit cost of hours for Process-1.
- c_3: Unit cost of hours for Process-2.
- c_4: Unit cost of hours for Process-3.
- H: Total hours available for all processes.
- t_a: Testing and inspection cost for Product-a (\$ unit).
- t_b: Testing and inspection cost for Product-b (\$ unit).
- t_c: Testing and inspection cost for Product-c (\$ unit).

Table 1: Monthly resource requirements for *abc*-PPP

Item	Product-a	Product-b	Product-c	Unit cost (\$)	Available
Material-abc (units)	6	8	10	50	300
Process-1 (hrs.)	20	25	28	30	700
Process-2 (hrs.)	12	15	18	15	600
Process-3 (hrs.)	8	10	14	10	500
Total time available (hrs.)					1600
Testing and inspection cost (\$)	50	75	100		
Planned demand (number)	23	18	20		
Selling price (\$)	1200	1600	2000		

Stochastic Parameters:

- $d_a(\omega)$: Demand for Product-a (units).
- $d_b(\omega)$: Demand for Product-b (units).
- $d_c(\omega)$: Demand for Product-c (units).

Table 2: Product demand scenarios

s	Scenario ω^s	Demand $(d_a(\omega^s), d_b(\omega^s), d_c(\omega^s))$	Probability $p(\omega^s)$	Description
1	ω^1	(15, 10, 5)	0.10	*Low*
2	ω^2	(20, 15, 15)	0.30	*Moderate*
3	ω^3	(25, 20, 25)	0.30	*High*
4	ω^4	(30, 25, 30)	0.20	*Extreme*
5	ω^5	(10, 10, 10)	0.10	*Rare*

1 Problem 3.1 (Risk-Neutral SLP with Recourse)

(a)

$$
\begin{aligned}
z^*_{RP} := \max \quad & \sum_{s \in \mathcal{S}} p^s \left(1150y_1^s + 1525y_2^s + 1900y_3^s\right) - (50x_1 + 30x_2 + 15x_3 + 10x_4) \\
\text{s.t.} \quad & -x_1 \geq -300 \\
& -x_2 \geq -700 \\
& -x_3 \geq -600 \\
& -x_4 \geq -500 \\
& -x_2 - x_3 - x_4 \geq -1600 \\
& x_1 - 6y_1^s - 8y_2^s - 10y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_2 - 20y_1^s - 25y_2^s - 28y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_3 - 12y_1^s - 15y_2^s - 18y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_4 - 8y_1^s - 10y_2^s - 14y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& -y_1^s \geq -d_a^s && \forall s \in \mathcal{S} \\
& -y_2^s \geq -d_b^s && \forall s \in \mathcal{S} \\
& -y_3^s \geq -d_c^s && \forall s \in \mathcal{S} \\
& x_1, x_2, x_3, x_4 \geq 0 \\
& y_1^s, y_2^s, y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& s \in S = \{1, 2, 3, 4, 5\}
\end{aligned}
$$

(b)

Table 3: Risk-Neutral SLP with Recourse Solution

Resource	**Decision**	**Amount**	**Scenario** s				
			Low (1)	Moderate (2)	High (3)	Extreme (4)	Rare (5)
Material-abc (units)	x_1	236.40					
Process-1 (hrs)	x_2	690.00					
Process-2 (hrs)	x_3	432.00					
Process-3 (hrs)	x_4	318.00					
Product-a	y_1^s		15.00	0	0	0	8.00
Product-b	y_2^s		10.00	10.80	10.80	10.80	10.00
Product-c	y_3^s		5.00	15.00	15.00	15.00	10.00
z^*_{RP} (Profit)		**2341.00**					

2 Problem 3.2 (Deterministic Models)

(a) Expected Value Solution

$$
\begin{aligned}
z^*_{EV} := \max \quad & (1150y_1 + 1525y_2 + 1900y_3) - (50x_1 + 30x_2 + 15x_3 + 10x_4) \\
\text{s.t.} \quad & -x_1 \geq -300 \\
& -x_2 \geq -700 \\
& -x_3 \geq -600 \\
& -x_4 \geq -500 \\
& -x_2 - x_3 - x_4 \geq -1600 \\
& x_1 - 6y_1 - 8y_2 - 10y_3 \geq 0 \\
& x_2 - 20y_1 - 25y_2 - 28y_3 \geq 0 \\
& x_3 - 12y_1 - 15y_2 - 18y_3 \geq 0 \\
& x_4 - 8y_1 - 10y_2 - 14y_3 \geq 0 \\
& -y_1 \geq -22 \\
& -y_2 \geq -17.5 \\
& -y_3 \geq -19.5 \\
& x_1, x_2, x_3, x_4, y_1, y_2, y_3 \geq 0
\end{aligned}
$$

Table 4: Expected Value Problem Solution

Resource	**Decision**	**Amount**
Material-abc (units)	x_1	244.28
Process-1 (hrs)	x_2	700.00
Process-2 (hrs)	x_3	443.40
Process-3 (hrs)	x_4	334.60
Product-a	y_1	0.00
Product-b	y_2	6.16
Product-c	y_3	19.50
z^*_{EV} (Profit)		**3233.00**

(b) Extreme Event Solution

$$
\begin{aligned}
z^{EES} := \max \quad & (1150y_1 + 1525y_2 + 1900y_3) - (50x_1 + 30x_2 + 15x_3 + 10x_4) \\
\text{s.t.} \quad & -x_1 \geq -300 \\
& -x_2 \geq -700 \\
& -x_3 \geq -600 \\
& -x_4 \geq -500 \\
& -x_2 - x_3 - x_4 \geq -1600 \\
& x_1 - 6y_1 - 8y_2 - 10y_3 \geq 0 \\
& x_2 - 20y_1 - 25y_2 - 28y_3 \geq 0 \\
& x_3 - 12y_1 - 15y_2 - 18y_3 \geq 0 \\
& x_4 - 8y_1 - 10y_2 - 14y_3 \geq 0 \\
& -y_1 \geq -15 \\
& -y_1 \geq -20 \\
& -y_1 \geq -25 \\
& -y_1 \geq -30 \\
& -y_1 \geq -10 \\
& -y_2 \geq -10 \\
& -y_2 \geq -15 \\
& -y_2 \geq -20 \\
& -y_2 \geq -25 \\
& -y_2 \geq -10 \\
& -y_3 \geq -5 \\
& -y_3 \geq -15 \\
& -y_3 \geq -25 \\
& -y_3 \geq -30 \\
& -y_3 \geq -10 \\
& x_1, x_2, x_3, x_4, y_1, y_2, y_3 \geq 0
\end{aligned}
$$

Table 5: Extreme Event Problem Solution

Resource	**Decision**	**Amount**
Material-abc (units)	x_1	130.00
Process-1 (hrs)	x_2	390.00
Process-2 (hrs)	x_3	240.00
Process-3 (hrs)	x_4	170.00
Product-a	y_1	0.00
Product-b	y_2	10.00
Product-c	y_3	5.00
z^*_{EES} (Profit)		**1250.00**

(c) Scenario Analysis

$$
\begin{aligned}
z_s^{SA} := \max \quad & (1150y_1^s + 1525y_2^s + 1900y_3^s) - (50x_1^s + 30x_2^s + 15x_3^s + 10x_4^s) \\
\text{s.t.} \quad & -x_1^s \geq -300 \\
& -x_2^s \geq -700 \\
& -x_3^s \geq -600 \\
& -x_4^s \geq -500 \\
& -x_2^s - x_3^s - x_4^s \geq -1600 \\
& x_1^s - 6y_1^s - 8y_2^s - 10y_3^s \geq 0 \\
& x_2^s - 20y_1^s - 25y_2^s - 28y_3^s \geq 0 \\
& x_3^s - 12y_1^s - 15y_2^s - 18y_3^s \geq 0 \\
& x_4^s - 8y_1^s - 10y_2^s - 14y_3^s \geq 0 \\
& -y_1^s \geq -d_a^s \\
& -y_2^s \geq -d_b^s \\
& -y_3^s \geq -d_c^s \\
& x_1^s, x_2^s, x_3^s, x_4^s, y_1^s, y_2^s, y_3^s \geq 0
\end{aligned}
$$

Table 6: Scenario-wise SA Problem Solution

Item	**Decision**	*Low (1)*	*Moderate (2)*	*High (3)*	*Extreme (4)*	*Rare (5)*
Material-abc (units)	x_1^s	130.0	239.6	250.0	250.0	180.0
Process-1 (hrs)	x_2^s	390.0	700.0	700.0	700.0	530.0
Process-2 (hrs)	x_3^s	240.0	438.0	450.0	450.0	330.0
Process-3 (hrs)	x_4^s	170.0	322.0	350.0	350.0	240.0
Product-a	y_1^s	0.0	0.0	0.0	0.0	0.0
Product-b	y_2^s	10.0	11.2	0.0	0.0	10.0
Product-c	y_3^s	5.0	15.0	25.0	25.0	10.0
z_s^{SA} (Objective)		1250.00	2810.00	3750.00	3750.00	2000.00

Expected value: $\mathbb{E}[z_s^{SA}] = \mathbf{3043.00}$

(d)

Table 7: Expected profit associated with each model's solution

Model		**Profit suggested by model (\$)**	**Expected profit (\$)**
RP			2341.00
EV		3233.00	2287.50
EES		1250.00	1250.00
SA		3043.00	
Scenario s			
Low	1	1250.00	1250.00
Moderate	2	2810.00	2296.50
High	3	3750.00	2276.50
Extreme	4	3750.00	2276.50
Rare	5	2000.00	1855.00

$$
\begin{aligned}
EVPI := z^{RP} - z^{WS} = z^{RP} - \sum_{s=1}^{5} p_s z_s^{SA} & \\
= \$2341.00 - \$3043.00 & \\
= -\$702.00 & \\
VVS = \$2341.00 - \$2287.50 & \\
= \$53.5 &
\end{aligned}
$$

It is necessary to use the RP model in this case, as the decision maker is willing to pay \$702 for perfect information on the product demand forecast and the value of the stochatic solution is \$53.5.

3 Problem 3.3 (Excess Probability)

$$
\begin{aligned}
\max \quad & \sum_{s \in \mathcal{S}} p^s \left(1150 y_1^s + 1525 y_2^s + 1900 y_3^s\right) - \left(50 x_1 + 30 x_2 + 15 x_3 + 10 x_4\right) - \lambda \sum_{s \in \mathcal{S}} p^s \gamma^s \\
\text{s.t.} \quad & -x_1 \geq -300 \\
& -x_2 \geq -700 \\
& -x_3 \geq -600 \\
& -x_4 \geq -500 \\
& -x_2 - x_3 - x_4 \geq -1600 \\
& x_1 - 6y_1^s - 8y_2^s - 10y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_2 - 20y_1^s - 25y_2^s - 28y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_3 - 12y_1^s - 15y_2^s - 18y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_4 - 8y_1^s - 10y_2^s - 14y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& -y_1^s \geq -d_a^s && \forall s \in \mathcal{S} \\
& -y_2^s \geq -d_b^s && \forall s \in \mathcal{S} \\
& -y_3^s \geq -d_c^s && \forall s \in \mathcal{S} \\
& -50x_1 - 30x_2 - 15x_3 - 10x_4 + 1150y_1^s + 1525y_2^s + 1900y_3^s + M\gamma^s \geq \eta && \forall s \in \mathcal{S} \\
& x_1, x_2, x_3, x_4, y_1^s, y_2^s, y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& \gamma^s \in \{0, 1\} && \forall s \in \mathcal{S}
\end{aligned}
$$

Table 8: MR-SLP with EP solution

Resource	**Decision**	**Amount**	**Scenarios s** *Low*	*Moderate*	*High*	*Extreme*	*Rare*
(a) $\eta := 0$ and $\lambda := 1000$							
Material-abc (units)	x_1	236.4					
Process-1 (hrs)	x_2	690.0					
Process-2 (hrs)	x_3	432.0					
Process-3 (hrs)	x_4	318.0					
Product-a	y_1		15	0	0	0	8
Product-b	y_2		10	10.8	10.8	10.8	10
Product-c	y_3		5	15	15	15	10
Scenario variable	γ^s		1	0	0	0	0
Objective value		2241.0					
Expected Profit		2341.0					
(b) $\eta := 2341.0$ and $\lambda := 1000$							
Material-abc (units)	x_1	236.4					
Process-1 (hrs)	x_2	690.0					
Process-2 (hrs)	x_3	432.0					
Process-3 (hrs)	x_4	318.0					
Product-a	y_1		15	0	0	0	8
Product-b	y_2		10	10.8	10.8	10.8	10
Product-c	y_3		5	15	15	15	10
Scenario variable	γ^s		1	0	0	0	1
Objective value		2141.0					
Expected Profit		2341.0					
(c) $\lambda := 0$ and Risk-Neutral RP							
Material-abc (units)	x_1	236.4					
Process-1 (hrs)	x_2	690.0					
Process-2 (hrs)	x_3	432.0					
Process-3 (hrs)	x_4	318.0					
Product-a	y_1		15	0	0	0	8
Product-b	y_2		10	10.8	10.8	10.8	10
Product-c	y_3		5	15	15	15	10
Scenario variable	γ^s		1	1	1	1	1
Objective value		2341.00					
Expected Profit		2341.00					

It is observable that the objective value of when $\eta := 2341$ is lower compared to when $\eta := 0$. This is because more scenarios were violated as the threshold increased. Yet, the expected profit of all three cases are the same. This implies that the penalty factor regarding risks is not significant enough to change decision-making.

4 Problem 3.4 (Conditional Value-at-Risk) - idk

$$
\begin{aligned}
\max \quad & \sum_{s\in\mathcal{S}} p^s \left(1150y_1^s + 1525y_2^s + 1900y_3^s\right) - (50x_1 + 30x_2 + 15x_3 + 10x_4) - \lambda\eta - \frac{\lambda}{1-\alpha}\sum_{s\in\mathcal{S}} p^s \nu^s \\
\text{s.t.} \quad & -x_1 \geq -300 \\
& -x_2 \geq -700 \\
& -x_3 \geq -600 \\
& -x_4 \geq -500 \\
& -x_2 - x_3 - x_4 \geq -1600 \\
& x_1 - 6y_1^s - 8y_2^s - 10y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_2 - 20y_1^s - 25y_2^s - 28y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_3 - 12y_1^s - 15y_2^s - 18y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_4 - 8y_1^s - 10y_2^s - 14y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& -y_1^s \geq -d_a^s && \forall s \in \mathcal{S} \\
& -y_2^s \geq -d_b^s && \forall s \in \mathcal{S} \\
& -y_3^s \geq -d_c^s && \forall s \in \mathcal{S} \\
& -50x_1 - 30x_2 - 15x_3 - 10x_4 + 1150y_1^s + 1525y_2^s + 1900y_3^s - \eta + \nu^s \geq 0 && \forall s \in \mathcal{S} \\
& x_1, x_2, x_3, x_4, y_1^s, y_2^s, y_3^s, \nu^s \geq 0, \eta \text{ unrestricted} && \forall s \in \mathcal{S}
\end{aligned}
$$

Table 9: MR-SLP with CVaR solution

Resource	Decision	Amount	Scenarios s *Low*	*Moderate*	*High*	*Extreme*	*Rare*
(a) $\alpha := 0.85$ and $\lambda := 100$							
Material-abc (units)	x_1	130.0					
Process-1 (hrs)	x_2	390.0					
Process-2 (hrs)	x_3	240.0					
Process-3 (hrs)	x_4	170.0					
Product-a	y_1		0	0	0	0	0
Product-b	y_2		10	10	10	10	10
Product-c	y_3		5	5	5	5	5
Scenario variable	ν^s		0	0	0	0	0
VaR	η	-1250					
Objective value		126250					
Expected Profit		1250					
(b) $\alpha := 0.95$ and $\lambda := 100$							
Material-abc (units)	x_1	130.0					
Process-1 (hrs)	x_2	390.0					
Process-2 (hrs)	x_3	240.0					
Process-3 (hrs)	x_4	170.0					
Product-a	y_1		0	0	0	0	0
Product-b	y_2		10	10	10	10	10
Product-c	y_3		5	5	5	5	5
Scenario variable	ν^s		0	0	0	0	0
VaR	η	-1250					
Objective value		126250					
Expected Profit		1250					
(c) $\lambda := 0$ and Risk-Neutral RP							
Material-abc (units)	x_1	236.4					
Process-1 (hrs)	x_2	690.0					
Process-2 (hrs)	x_3	432.0					
Process-3 (hrs)	x_4	318.0					
Product-a	y_1		15	0	0	0	8
Product-b	y_2		10	10.8	10.8	10.8	10
Product-c	y_3		5	15	15	15	10
Scenario variable	ν^s		0	0	0	0	0
VaR	η	-					
Objective value		2341.00					
Expected Profit		2341.00					

It is shown that all scenarios of the first and second model yield the same profit, so the CVaR equals the VaR. Since there is no variation across scenarios, changing α does not affect decisions or the objective value. Yet, it is observable that the objective values of the risk-averse approaches are lower than the risk-neutral approach, because decisions in risk-averse models are made in a relatively conservative way as means of considering risk matters - compared to a risk-neutral model.

5 Problem 3.5 (Expected Excess)

$$
\begin{aligned}
\max \quad & \sum_{s \in \mathcal{S}} p^s \left(1150 y_1^s + 1525 y_2^s + 1900 y_3^s\right) - (50x_1 + 30x_2 + 15x_3 + 10x_4) - \lambda \sum_{s \in \mathcal{S}} p^s \nu^s \\
\text{s.t.} \quad & -x_1 \geq -300 \\
& -x_2 \geq -700 \\
& -x_3 \geq -600 \\
& -x_4 \geq -500 \\
& -x_2 - x_3 - x_4 \geq -1600 \\
& x_1 - 6y_1^s - 8y_2^s - 10y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_2 - 20y_1^s - 25y_2^s - 28y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_3 - 12y_1^s - 15y_2^s - 18y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_4 - 8y_1^s - 10y_2^s - 14y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& -y_1^s \geq -d_a^s && \forall s \in \mathcal{S} \\
& -y_2^s \geq -d_b^s && \forall s \in \mathcal{S} \\
& -y_3^s \geq -d_c^s && \forall s \in \mathcal{S} \\
& -50x_1 - 30x_2 - 15x_3 - 10x_4 + 1150y_1^s + 1525y_2^s + 1900y_3^s + \nu^s \geq \eta && \forall s \in \mathcal{S} \\
& x_1, x_2, x_3, x_4, y_1^s, y_2^s, y_3^s, \nu^s \geq 0 && \forall s \in \mathcal{S}
\end{aligned}
$$

Table 10: MR-SLP with EE solution

			Scenarios s				
Resource	**Decision**	**Amount**	*Low*	*Moderate*	*High*	*Extreme*	*Rare*
(a) $\eta := 0$ and $\lambda := 1000$							
Material-abc (units)	x_1	234.5					
Process-1 (hrs)	x_2	690.0					
Process-2 (hrs)	x_3	429.8					
Process-3 (hrs)	x_4	312.9					
Product-a	y_1		15	0	0	0	8
Product-b	y_2		10	12.9	12.9	12.9	10
Product-c	y_3		5	13.2	13.2	13.2	10
Scenario variable	ν^s		0	0	0	0	0
Objective value		2238.9					
Expected Profit		2238.9					
(b) $\eta := 2341.0$ and $\lambda := 1000$							
Material-abc (units)	x_1	226.1					
Process-1 (hrs)	x_2	672.0					
Process-2 (hrs)	x_3	415.9					
Process-3 (hrs)	x_4	298.5					
Product-a	y_1		14.1	0	0	0	7.1
Product-b	y_2		10	15	15	15	10
Product-c	y_3		5	10.6	10.6	10.6	10
Scenario variable	ν^s		2063.4	0	0	0	613.4
Objective value		−265613.4					
Expected Profit		2073.3					
(c) $\eta := 0$ and $\lambda := 0$ and Risk-Neutral RP							
Material-abc (units)	x_1	236.4					
Process-1 (hrs)	x_2	690.0					
Process-2 (hrs)	x_3	432.0					
Process-3 (hrs)	x_4	318.0					
Product-a	y_1		15	0	0	0	8
Product-b	y_2		10	10.8	10.8	10.8	10
Product-c	y_3		5	15	15	15	10
Scenario variable	ν^s		180	0	0	0	0
Objective value		2341.00					
Expected Profit		2341.00					

As the threshold increases, the model shows more conservative decision-making behavior, as seen in the comparison between case (a) and case (b). While the risk-neutral approach aims to maximize the expected profit, risk-averse approaches adjust decisions to mitigate the probability of earning below the specified profit threshold.

6 Problem 3.6 (Probabilistically Constrained SP)

$$
\begin{aligned}
\max \quad & \sum_{s\in\mathcal{S}} p^s\left(1150y_1^s + 1525y_2^s + 1900y_3^s\right) - (50x_1 + 30x_2 + 15x_3 + 10x_4) & \\
\text{s.t.} \quad & -x_1 \geq -300 & \\
& -x_2 \geq -700 & \\
& -x_3 \geq -600 & \\
& -x_4 \geq -500 & \\
& -x_2 - x_3 - x_4 \geq -1600 & \\
& x_1 - 6y_1^s - 8y_2^s - 10y_3^s \geq 0 & \forall s\in\mathcal{S} \\
& x_2 - 20y_1^s - 25y_2^s - 28y_3^s \geq 0 & \forall s\in\mathcal{S} \\
& x_3 - 12y_1^s - 15y_2^s - 18y_3^s \geq 0 & \forall s\in\mathcal{S} \\
& x_4 - 8y_1^s - 10y_2^s - 14y_3^s \geq 0 & \forall s\in\mathcal{S} \\
& -y_1^s + M^s z^s \geq -d_a^s & \forall s\in\mathcal{S} \\
& -y_2^s + M^s z^s \geq -d_b^s & \forall s\in\mathcal{S} \\
& -y_3^s + M^s z^s \geq -d_c^s & \forall s\in\mathcal{S} \\
& 0.1z^1 + 0.3z^2 + 0.3z^3 + 0.2z^4 + 0.1z^5 \leq 1-\alpha & \forall s\in\mathcal{S} \\
& x_1, x_2, x_3, x_4, y_1^s, y_2^s, y_3^s, \nu^s \geq 0, z^s \in \{0,1\} & \forall s\in\mathcal{S}
\end{aligned}
$$

Table 11: MR-SLP with Probabilistically Constrained solution

Resource	**Decision**	**Amount**	**Scenarios s** *Low*	*Moderate*	*High*	*Extreme*	*Rare*
(a) $\alpha := 0.9$							
Material-abc (units)	x_1	250.0					
Process-1 (hrs)	x_2	700.0					
Process-2 (hrs)	x_3	450.0					
Process-3 (hrs)	x_4	350.0					
Product-a	y_1		0	0	0	0	8.5
Product-b	y_2		0	11.2	0	0	10
Product-c	y_3		25	15	25	25	10
Scenario variable	z^s		1	0	0	0	0
Objective value		2826.5	Expected Profit		2826.5		
(b) $\alpha := 0.8$							
Material-abc (units)	x_1	250.0					
Process-1 (hrs)	x_2	700.0					
Process-2 (hrs)	x_3	450.0					
Process-3 (hrs)	x_4	350.0					
Product-a	y_1		0	0	0	0	0
Product-b	y_2		10	10	10	10	10
Product-c	y_3		5	5	5	5	5
Scenario variable	z^s		1	0	0	0	1
Objective value		3174.0					
Expected Profit		3174.0					
(c) Risk-Neutral RP							
Material-abc (units)	x_1	236.4					
Process-1 (hrs)	x_2	690.0					
Process-2 (hrs)	x_3	432.0					
Process-3 (hrs)	x_4	318.0					
Product-a	y_1		15	0	0	0	8
Product-b	y_2		10	10.8	10.8	10.8	10
Product-c	y_3		5	15	15	15	10
Scenario variable	z^s		0	0	0	0	0
Objective value		2341.00					
Expected Profit		2341.00					

Compared to (b), model (a) takes more conservative decisions due to a stricter probabilistic constraint, which forces the model to satisfy scenario constraints in a higher proportion of cases. In contrast, the risk-neutral model (c) does not allow for violation, resulting in more conservative decisions overall. However, both (a) and (b) strategically relax a small subset of scenarios using probabilistic constraints, enabling them to achieve higher expected profits than the risk-neutral baseline by trading off feasibility in few cases.

7 Problem 3.7 (Quantile Deviation) - idk

$$
\begin{aligned}
\max \quad & \sum_{s\in\mathcal{S}} p^s\left(1150y_1^s + 1525y_2^s + 1900y_3^s\right) - (50x_1 + 30x_2 + 15x_3 + 10x_4) - \lambda\left[\epsilon_1 \sum_{s\in\mathcal{S}} p^s u^s + \epsilon_2 \sum_{s\in\mathcal{S}} p^s v^s\right] \\
\text{s.t.} \quad & -x_1 \geq -300 \\
& -x_2 \geq -700 \\
& -x_3 \geq -600 \\
& -x_4 \geq -500 \\
& -x_2 - x_3 - x_4 \geq -1600 \\
& x_1 - 6y_1^s - 8y_2^s - 10y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_2 - 20y_1^s - 25y_2^s - 28y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_3 - 12y_1^s - 15y_2^s - 18y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& x_4 - 8y_1^s - 10y_2^s - 14y_3^s \geq 0 && \forall s \in \mathcal{S} \\
& -y_1^s \geq -d_a^s && \forall s \in \mathcal{S} \\
& -y_2^s \geq -d_b^s && \forall s \in \mathcal{S} \\
& -y_3^s \geq -d_c^s && \forall s \in \mathcal{S} \\
& u^s + (1150y_1^s + 1525y_2^s + 1900y_3^s - 50x_1 - 30x_2 - 15x_3 - 10x_4) - \eta \geq 0 && \forall s \in \mathcal{S} \\
& v^s - (1150y_1^s + 1525y_2^s + 1900y_3^s - 50x_1 - 30x_2 - 15x_3 - 10x_4) + \eta \geq 0 && \forall s \in \mathcal{S} \\
& x_1, x_2, x_3, x_4, y_1^s, y_2^s, y_3^s, u^s, v^s \geq 0, \eta \text{ unrestricted} && \forall s \in \mathcal{S}
\end{aligned}
$$

Table 12: MR-SLP with QDEV solution

			Scenarios s				
Resource	**Decision**	**Amount**	*Low*	*Moderate*	*High*	*Extreme*	*Rare*
(a) $\alpha := 0.85, \epsilon_1 = 0.3, \epsilon_2 = 0.17, \lambda = 5.88$							
Material-abc (units)	x_1	220.0					
Process-1 (hrs)	x_2	655.0					
Process-2 (hrs)	x_3	405.0					
Process-3 (hrs)	x_4	290.0					
Product-a	y_1		13.25	0	0	0	6.3
Product-b	y_2		10	15	15	15	10
Product-c	y_3		5	10	10	10	10
Scenario variable	u^s		1887.5	0	0	0	437.5
Scenario variable	v^s		0	0	0	0	0
Eta	η	2250.0					
Objective value		2017.5					
Expected Profit		1607.2					
(b) $\alpha := 0.95, \epsilon_1 = 0.1, \epsilon_2 = 0.19, \lambda = 5.26$							
Material-abc (units)	x_1	236.4					
Process-1 (hrs)	x_2	690.0					
Process-2 (hrs)	x_3	432.0					
Process-3 (hrs)	x_4	318.0					
Product-a	y_1		15	0	0	0	8
Product-b	y_2		10	10.8	10.8	10.8	10
Product-c	y_3		5	15	15	15	10
Scenario variable	u^s		2970	0	0	0	1520
Scenario variable	v^s		0	0	0	0	0
Eta	η	2790.0					
Objective value		2104.7					
Expected Profit		2341.0					
(c) $\lambda := 0$ and Risk-Neutral RP							
Material-abc (units)	x_1	236.4					
Process-1 (hrs)	x_2	690.0					
Process-2 (hrs)	x_3	432.0					
Process-3 (hrs)	x_4	318.0					
Product-a	y_1		15	0	0	0	8
Product-b	y_2		10	10.8	10.8	10.8	10
Product-c	y_3		5	15	15	15	10
Objective value		2341.00					
Expected Profit		2341.00					

Compared to (b), model (a) uses a smaller α, which leads to a lower quantile reference point η, resulting in more conservative decisions and reduced expected profit. Model (a) takes more conservative decisions than Model (c) due to the QDEV-based penalty, which discourages profit falling below a certain quantile threshold. Yet, it is notable that Models (b) and (c) yield the same solution because the QDEV penalty, although triggered, was not significant enough to affect the optimal decision.

8 Problem 3.11 (Risk Measure)

8.1 Comparision between Risk-Averse Measures

Quantile-based risk measures (e.g., EP, QDEV, CVaR) focus on limiting or penalizing outcomes in the worst α-quantile of the profit distribution. These methods assess tail risk by imposing penalties based on how often (EP), how far (QDEV), or how much on average (CVaR) outcomes fall below a specified quantile threshold. Deviation-based risk measures (e.g., EE, ASD, CDEV) penalize deviations from a central reference, such as the mean or a fixed profit threshold. They aim to reduce variability and promote stability across scenarios, even if the worst-case scenarios are not explicitly targeted.

8.2 Risk-Neutral vs Risk-Averse: Application

The risk-neutral solution is appropriate when the decision-maker is focused solely on maximizing expected value and is indifferent to variability or downside risk. This is suitable in applications when failures carry minimal consequences. For instance, in inventory management for low-cost consumer goods, occasional stockouts may be tolerable if the expected profit is maximized in the long run. In contrast, the risk-averse solution is preferred when uncertainty and downside risk significantly impact the stakeholder. For example, in energy managment systems, where supply shortages can cause blackouts, or in financial portfolio optimization, where losses in rare events must be controlled, quantile-based measures like CVaR or QDEV are appropriate for limiting tail risk.